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Common User Interface (CUI) Programme: Clinical Applications and Patient Safety (CAPS) Project: Development Lifecycle

Amendment History:

Version	Date	Amendment History
0.1	10-Apr-2006	Barry Day - First draft for comment
0.2	22-May-2006	Barry Day - Second draft, incorporating comments from Roarke Batten and ISB.
0.3	19-Jan-2007	Third draft, incorporating comments from Roarke Batten, Tim Chearman and ISB.
0.4	23-Jan-2007	Roarke Batten - updated following discussion with Tim and Bev.
0.5	23-Jan-2007	Roarke Batten - updated following final review with Bev Scott prior to the ISB January submissions meetings. For confirmation of sponsorship message from Mark Ferrar, and review and approval of the overall document content by Mark Ferrar.
0.5	22-Mar-2007	Roarke Batten - updates to the document following ISB feedback in January submissions. This version maintained as 0v5 to avoid removing and updating references to final release documentation in all submissions for January and February.
1v1	19-Sep-2007	Roarke Batten – updates reflecting clarity of the definition of “clinical systems” earlier in the document, updates with changed workstream names in Phase 2, and updated sponsor statement from Mark Ferrar.
2v0	24-Oct-2007	Roarke Batten – second baseline document for publication.
2v1	12-Feb-2008	Roarke Batten – clinical systems definition for the purposes of scoping for the ISB HaSC included following ISB HaSC appraisal.
3v0	15-Feb-2008	Roarke Batten – third baseline version approved for use in relation to ISB publications.
4v0	04-Aug-2009	Bev Scott – fourth baseline version. Updated to incorporate change in directorate for CUI CAPS project from Infrastructure to Data Standards and Products. Also change in organisation from NHS Connecting for Health to the Department of Health Informatics Directorate.
5v0	26-Aug-2009	Bev Scott – fifth baseline version. Updated with the CUI CAPS Clinical Safety Management System and the UX development lifecycle.

Reviewers:

This original document has been reviewed by the following Beverley Scott, Martin Severs, Maureen Baker, Ian Harrison, Paul Hough, Mark Ferrar and DHID CAPS Team members as described in this document.

Approvals:

This document requires the following approvals.

Name	Signature	Title	Date of Issue	Version
Roarke Batten	Pending approval	CUI Programme Manager	30-Oct-2007	2v0
Roarke Batten	Pending approval	CUI Programme Manager	15-Feb-2008	3v0
Roarke Batten	Pending approval	CUI Programme Manager	04-Aug-2009	4v0

Related Documents:

These documents (some of which are confidential) have been used in the production of this lifecycle.

Ref no	Doc Reference Number	Title	Version
1	NPFIT-SHR-QMS-PRP-0015	Glossary of Terms Consolidated.doc	<enter latest>
2	PID draft 0v8 (NPFIT-FNT-TO-DPM-CUI0001) - all amendments included[1][1].doc	Project Initiation Document	0v8
3	NPFIT-FNT-TO-TOCLNSA-0059 (Request for Information)	Authority Advisory Note: Contractor Clinical Safety Management Systems	1.0
4	requirements_template.doc	ISB HaSC Operational Standard 'Requirement' level submission template	10.0
5	draft_template.doc	ISB HaSC Operational Standard 'Draft' level submission template	10.0
6	full_template.doc	ISB HaSC Operational Standard 'Full' level submission template	10.0

Glossary of Terms

Term	Acronym	Definition
Information Standards Board for Health and Social Care	ISB HaSC	Independent body responsible for approving information standards for health and social care.
Information Standards Board	ISB	Former title of the ISB HaSC
Microsoft	n/a	Microsoft Corporation®, and its subsidiaries
Microsoft Solutions Framework	MSF	Service delivery framework developed and used by Microsoft
Patient Safety Assessment	PSA	A hazard assessment used to assess clinical safety
Statement of Work	SoW	An output of the MSF defining key pieces of work to be delivered over a release cycle.
Data Standards and Products	DS&P	Directorate within Department of Health Informatics Directorate responsible for delivery of the CUI CAPS Project
Department of Health Informatics Directorate	DHID	New organisation following restructure of NHS Connecting for Health

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Purpose of the document

This document outlines the delivery lifecycle applicable to the work of the Clinical Applications and Patient Safety (CAPS) Project, which forms a part of the Department of Health Informatics Directorate (DHID), Common User Interface (CUI) programme. The lifecycle intends to draw together the multiple stakeholders involved in the work of CAPS and identify interdependencies between them. In particular, this document describes the integration of NHS Clinical Safety and the NHS Information Standards activities, in order to satisfy both safety requirements, authorities, and to gain successful standards submissions through the Information Standards Board for Health and Social Care (ISB HaSC).

The document also serves as a supplementary source of generic process information, in support of each of the standards submitted to the ISB HaSC, where a single source of generic cross-standard information better suits the purposes of submissions, than duplication of information across the submissions.

All new versions of this document will be referenced in subsequent submissions to the ISB HaSC. Section 4 of this document contains the information necessary to support the Evaluation requirements of an ISB HaSC submission.

Intended Audience

This document is primarily aimed at a group of those stakeholders who currently interconnect with the CAPS Project, including the ISB HaSC, DHID Clinical Safety Group (CSG), the NHS National Integration Centre (NIC), Local Service Providers (LSPs), National Application Service Providers (NASPs) Independent Software Vendors (ISVs), and Microsoft.

It is assumed, therefore, that the representatives to whom this document is circulated to will have knowledge of the NHS, Department of Health Informatics Directorate (DHID), the history of the National Programme for IT (NPfIT), impact of the National Local Ownership Programme (NLOP) and the Regional Programmes for IT that have evolved from this, as well as the high-level organisational structure of the Department of Health (DH). Where relevant, references to related documents will be given for further reading.

Background to CUI

The CUI Programme is a funded programme within DHID. Phase 2 of the programme is split into five Projects:

1. Clinical Applications and Patient Safety (CAPS) [previously “Design Guide” and “Toolkit”]
2. Information Management and Technology (IM&T) Tools
3. Knowledge Worker Tools (KWT)
4. NHS Clinicians Portal
5. NHSmail Technical Refresh (MTR)

The lifecycle outlined in this document covers the design guidance outputs from the CAPS Project only.

The high-level objective of the CAPS Project is to establish user interface design guidelines, standards and associated technical components for the high-priority, safety-critical areas of clinical functionality contained, or planned to be contained, within clinical systems provided to the NHS in England.

The remit of the output from the CAPS Project within the CUI programme is described in each Standard Submission as "clinical systems within the NHS in England within the remit of NHS Connecting for Health".

Clinical systems within the context of the CAPS Project are defined as all those software applications within the contractual control of NHS Connecting for Health used in the processing of patient identifiable data. This includes those clinical systems delivered within the scope of the existing LSP and National Application Service Provider (NASP) contracts.

It is recommended that all other suppliers of clinical systems that include CUI standards move towards implementing them (once they are approved) as part of their updating and maintenance programmes.

For more information on the CUI programme, please contact the CUI team via cui stakeholder.mailbox@nhs.net or visit the programme website at www.cui.nhs.uk

Design Guidance Development Lifecycle

Overview

Figure 1 on the following page provides a diagram outlining the development lifecycle. Due to the size of the diagram it is best to read it in electronic format. It is possible, however, to make some general comments regarding the lifecycle, which are made below. A *.vsd file is provided here containing the diagram in order to enable detailed viewing.



CUI CAPS - UX
Development Lifecycle

Developed upon existing processes

The lifecycle consists of five stages: *Envisioning*, *Planning*, *Development*, *Stabilisation* and *Live*. These stages are derived from Microsoft's Solutions Framework (MSF), as much of the development of the design guidance (and solutions toolkit) will be carried out in partnership with Microsoft, the DHID development partner working on the CUI programme. Other established processes integrated into the lifecycle are derived as below:

- The Clinical / Patient Safety activities referenced in this document are taken from the Clinical Safety Management System as defined for suppliers of systems to the NHS under the NPfIT (see document references 1 and 2);
- The Information Standards process for developing standards is taken from existing templates for Operational Standards developed by the ISB HaSC (see document references 4-6).

Releases and work items

The CUI programme will run over two phases, the first of which was covered by the previous version of this lifecycle document. This version of the document has been updated for the second phase of the programme running from October 2007 until December 2009.

The second phase is broken down into several releases, of anything between four and nine months in duration, dependent on the needs of the work area being covered in a given release. This lifecycle document describes activities which occur in relation to a single release. Each CAPS release has several work items (eg. Medication Line, Prescribing, Date and Time controls, and a Clinical Terminology user interface) which may result in the production of a Design Guide entry.

The activities listed in the lifecycle are either applicable across all work items in a release, or a single work item. This differentiation is indicated by the 'light', 'medium' and 'heavy' rating respectively. In any one release it is likely that different work items will be at different stages of maturity. It is realistic, for example, to be developing a piece of design work on medications management (Development stage) whilst carrying out a pilot of Date and Time input controls implemented in a test system (Stabilisation stage).

Some of the larger areas of scope within the CAPS Project (for example, medications management) are likely to require multiple releases of development to achieve a level of maturity satisfactory to stakeholders and suitable to publish as a complete Design Guide entry.

The general landscape

The majority of activities are carried out by Microsoft and the DHID CAPS team. This is greatest between the Envisioning and Development stages. There is little activity from Microsoft in Stabilisation and Live once they have delivered the Design Guidance. In these stages the focus shifts to the DHID CAPS Project team and other key stakeholders.

In all stages there is an important integration with the Clinical Safety Group, who assures the clinical safety work being carried out. Similarly, The ISB HaSC assurance activities play a key role from Planning to Live assessing the rigour of the CAPS work as pieces of design guidance are submitted to become standards. Attaining Clinical Safety sign off on these work items is a precursor to delivering a Full Standard. Both the Clinical Safety Group and ISB HaSC have a key role to play during Stabilisation in determining the required level of implementation of a piece of design guidance in order to attain Draft and Full Standards approvals, explained in more detail later on.

The next section describes in more detail the activities at each stage.

Consultation and support

The following represents the CAPS Project team during Phase 2 of the CUI programme:

Role	Names
Programme Director	Ken Lunn
User Interface Design Strategy	Stephen Corbett
Programme Manager	Roarke Batten
Technical Project Leads	Tim Chearman, Mike Carey
User Experience Architect	Kit Lewis

Role	Names
Clinical Terminology Architect	Peter Johnson
Clinical Architect	Mike Bainbridge
Clinical Advisors	A varying team of clinical advisors and leads across NHS care settings such as Hospital Doctor, General Practitioner, Primary and Secondary care Nursing and Pharmacy.
Clinical Safety Advisor	Beverley Scott (February 2008 – September 2009) Lindsey Butler (August 2009 – onwards)

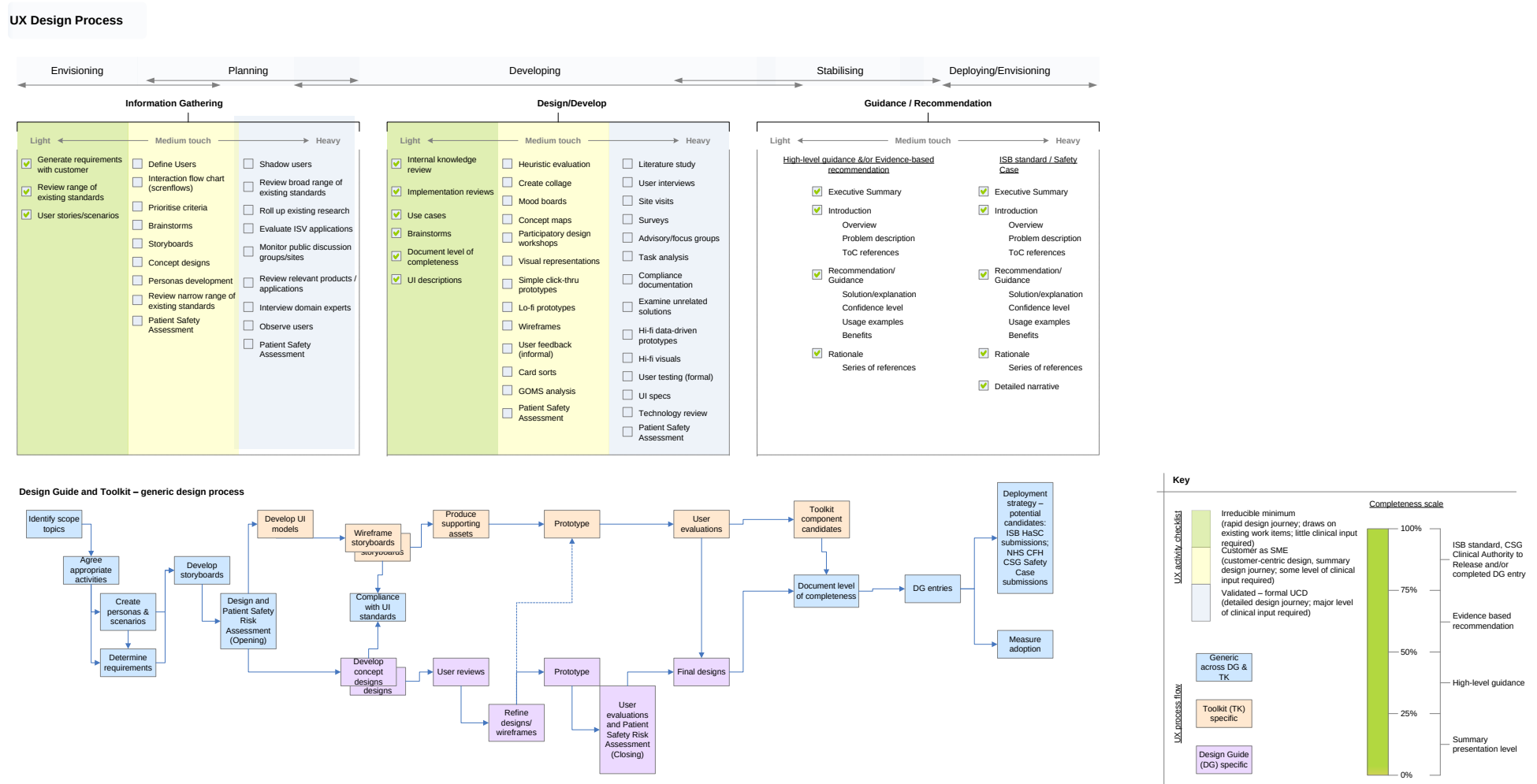
Input from specific clinical stakeholder bodies and required governance approval specific to each area of guidance under development within individual standards are detailed within individual standards and supporting documentation.

Consideration is given to the impact upon the patient experience of clinical systems used by clinicians, in terms of requirement, risks and hazards, during the various phases of clinical engagement where applicable to each standard.

In addition to specific sponsors and approvers, the support and sponsorship where applicable of each of the following bodies is also in place for guidance and related standards developed within this programme:

- The National Patient Safety Agency (NPSA) and DHID Clinical Safety Group in ratification and signoff of the clinical safety process against which each design entry is developed;
- Royal Colleges affected by each area of guidance, through consultation in the development of guidance during high-level review and signoff of related areas of guidance;
- The NHS, through direct engagement with NHS Professionals throughout user requirements gathering, user surveys and interviews, site visits, workshops, hazard assessments and user testing;
- The National Clinical Executive and the Office of the Chief Clinical Officer (OCCO) as sponsor, reviewer and approver of the table of contents and roadmap for the CAPS Project as well as high level signoff of related areas of guidance;
- The National Clinical Leads group (National Clinical Architects for the NPfIT and NLOP programmes) in evangelising and supporting the programme outputs and sourcing relevantly experienced clinical advisory support;

- The NHS Care Record Service National Clinical Reference Panel as a monitor of the work developed, reviewed and approved by the Design Steering Groups;
- Design Steering Groups constituting multiple practising clinical staff from across the NHS in areas of authority and experience, in user requirements analysis, reviews and hazard assessments;
- National Advisory Groups (NAGs), national forums representing clinical and health disciplines, presented with the outputs of the programme group meetings, as well as attending break out consultation workshops on the design guidance in development;
- The National Clinical Engagement team, Regional Clinical Engagement teams, and NHS Clinical Safety Team in sourcing a wide range of clinicians to be engaged in consultation, requirements gathering and user testing activities against the work during development and once complete;
- Various suppliers of clinical systems to the NHS under the contractual control of DHID have and will be actively involved in the development and implementation of the design guidance standards. This involvement is not however restricted to those suppliers providing services to the NHS directly. Engagement with other clinical software suppliers and / or healthcare procurement groups and agencies both in the UK and internationally has been and will continue to be actively undertaken to ensure the guidance produced is relevant and applicable to as broad a range of healthcare applications as possible.

Figure 1: DHID CUI CAPS User Experience (UX) Development Lifecycle ([PDF version](#))

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Sponsor statement

The following sponsor statement (via email on 4th August 2009) has been agreed by Kenn Lunn (Director of Data Standards and Products for the Technology Office and the CUI CAPS Clinical Safety Advisor).

As the Director of Data Standards and Products, I would like to confirm in writing my support for the package of standards that will be submitted to the ISB HaSC in relation design guidance developed within the Common User Interface (CUI) programme. This statement is in support of all design guidance developed and for submittal to the ISB HaSC between April 2009 and December 2009.

I can confirm the proposed standards fit with NHS policy drivers, NHS Business Architecture and NHS Data Standards.

Furthermore, I can assure that the conditions for achieving implementation and for maintenance thereof post implementation where required, including funding, are present. I also support the forecast dates for full compliance with these standards, as detailed in the individual submissions.”

Ken Lunn
Director of Data Standards and Products
Technology Office
Department of Health Informatics Directorate

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Compliance

Commercial Issues

The relevant Intellectual Property Rights, which might concern the use of the proposed standard within the NHS, are owned jointly by NHS Connecting for Health and Microsoft. This is defined in the Development Agreement, Clause 13, "Intellectual Property Rights".

The Development Agreement is an exhibit of the Contract between the Authority (NHS) and the Company (Microsoft), signed December 2005 and renewed in October 2007.

Design Guidance and Standards are available on a royalty free licence basis to any and all suppliers to the NHS in England, as well as to other national and international clinical systems suppliers and health procurement agencies, as long as such use is in line with the terms and conditions defined in the Participation Agreement (where suppliers or other agents take an active role in the development process) or in the End User Licence Agreement, the terms of which must be accepted prior to access being granted to the CUI web site where all guidance will be published.

The CUI development agreement with Microsoft will continue to provide support and development of guidance and standards to end December 2009. Ongoing maintenance costs in relation to guidance and standards will be provided for beyond this date within the operating budget of the DHID Technology Office.

Strategic fit

The CUI programme forms a component of the Technology Office strategy to deliver and support the goals and objectives of the NPfIT. The resultant CUI CAPS Information Standards will continue to support the delivery of this agreed national programme of work involving the necessary use of information technology, through a process of clinical systems review against the core objectives of the NPfIT and DHID:

- Increased patient safety;
- Increased clinical take up of systems;
- Increased clinician effectiveness and efficiency;
- Reduced training costs.

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Operational Fit

The proposed standards will cover the human interface of clinical systems only. They do not relate to, and as such will not cover underlying processing or data storage.

Operationally it is envisaged that clinical system developers will use the standards in two ways:

1. When evaluating an existing system against the standard, developers will make a direct comparison against the existing designs and that defined in the standards;
2. When commissioning and designing a new system the standard would be applied early in the user interface design process.

Currently all LSPs and NASPs are in the process of developing the systems which will enable access to electronic patient records. These systems may implement design solutions which diverge from the proposed standards. Please see sections **6.7 Migration Plans** and **7 Maintenance** for further detail on how a plan for this will be developed.

Interdependencies

There are a range of interdependencies which have been evaluated and documented for each standard. The CAPS team has reviewed relevant known standards. It is envisaged that none of those reviewed and listed in each submission will prevent progression of standards through the ISB HaSC process at this stage. In the event of issues arising, the CAPS team will work with the relevant body to resolve any issues. This may result in updates to CUI CAPS standards.

Governance

Identification of governance issues relating to clinical safety, patient confidentiality, data security, etc. will be included in hazard assessment workshops and safety reviews, as well as a part of the clinical user testing process.

Accredited clinicians will be used to either host or take part in hazard assessments against each of the clinical areas covered by the design guidance in development. This will ensure trained specialists are involved in the process, with expertise in the identification of these types of issue.

In addition to this, the design guide team will develop and review design guidance in development against all known and relevant bodies, codes of practice, and standards relevant to the clinical area being addressed. This will ensure adherence to the expected level of rigour within DHID programmes.

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Provisional testing approach

As outlined in the section regarding the general lifecycle each stage of development has been subject to iterative testing and validation from a clinical, user and patient safety perspective. An DHID Hazard Log is available for reference. This records patient safety risks and mitigations throughout the design process. In terms of user engagement, there are two levels.

During Development end users are involved in much of the user research and user testing undertaken. Additionally relevant Royal Colleges and Professional Regulatory bodies will be consulted across the CUI programme.

During the pilot stage further clinical user testing and patient safety analysis will be undertaken. The results will be used in analysis of the designs proposed and used in rework where applicable. This may also result in general recommendations for human behavioural, organisational and technical user guidance to be given by suppliers.

The NIC also plays a key role in the technical assurance of design guidance, and this activity begins at the end of Development and continues through to the Live stage. This is linked to system suppliers implementing design guidance, which occurs during Stabilisation and Live. The suppliers are also involved in the CAPS usability assessments which are carried out during Development.

Supplier testing will be carried out as part of their standard release process. This will cover clinical and functional testing. Specific human behavioural, organisational and technical user guidance will be given by the application supplier to their users as part of the standard release process for that supplier.

High level performance characteristics are specified within each standard as a first point of reference for testing criteria.

Compliance Testing

A set of compliance criteria will be developed to enable the NIC to assess adherence with the CUI CAPS standards across LSP hosted systems. These will be applied to systems as and when they enter the test environments in the NIC following the required compliance date for a given standard.

These tests will be objective, measurable and quantifiable, and will reflect the specific requirement of each given piece of design guidance. These tests will be based on a mix of software tests and human evaluations, as per existing NIC processes.

Compliance will be required for these systems based on the commitment to be compliant with all current data standards within the LSP contracts.

The compliance criteria will also be made available to any systems suppliers wishing to adopt the CAPS design guidance and not currently under the contractual control of

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DHID, as well as to health organisations and procurement agents wishing to ensure a level of adherence with the guidance within systems procured by them.

Migration Plans

Compliance dates defined for each standard submission relate to the dates by which the ISVs for each of the software applications which fall under the definition in section 3 above must comply with the CUI CAPS standards, as per relevant contractual requirements within the NPfIT and Regional Programmes for IT for supplier compliance with NHS Information Standards

For those systems covered by the DHID Existing Systems Programme, a standard compliance plan and strategy will be developed subsequent to the formal compliance dates defined above for LSP hosted systems.

This will be provided to each existing system provider on a case by case basis as and when they sign up to use DHID systems (i.e. access data on the SPINE or use a SPINE related service in their application, such as Choose and Book).

Maintenance

Once a Design Guide entry or a Full Information Standard is published in the public domain and is implemented by suppliers, a system will be in place to manage the ongoing ownership of and any arising problems with the standard.

There will be two forms of maintenance which are described below.

Reactive maintenance

During the life of the CUI programme it is possible that a supplier may contend the viability of a piece of CAPS guidance/standard, or may present an equally clinically safe alternative solution. It is important therefore that a feedback loop is set-up between compliance testing processes at the NIC and the CAPS team, where the CAPS team are consulted on either the clinical evidence of alternate designs or to provide feedback to suppliers on design rationale, if appropriate.

The CAPS Project will publish only a single standard for each aspect of the interface; however this may not be the best outcome for the NHS, so a review mechanism with links to the ISB HaSC will need to consider any such issues as they arise. Any significant changes to a standard will need assurance of the likes of that proposed in the above lifecycle.

There is currently a three yearly cycle of standards review undertaken by the ISB HaSC. Sufficient resource and funding will be available within the Technology Office, such that if review of any CUI CAPS Full Standard results in a need for rework of the standard, a work package can be defined and team assembled to ensure the work is completed to update the standard accordingly.

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For clinical incidents that occur within a trust that relate to a piece of CAPS design and which are reported through the supplier Clinical Safety Incident Report Register, a mechanism will be set-up to ensure that any design resolution involves CAPS. Any significant changes to a standard will need assurance of the likes of that proposed in the above lifecycle. This will be an enhancement to the existing Clinical Incident Reporting process defined by the Clinical Safety Group.

Proactive maintenance

In addition to the reactive approach above, CAPS is taking a proactive approach during the life of the CUI development programme to identifying issues with usability, interoperability, implementability and clinical safety by approaching system suppliers and system users for evidence. This may take the form of reviewing a supplier's hazard log or interviewing technical staff in a trust. There are many techniques for information extraction. All evidence gathered will then feed into a standards design review which will occur either during the 3 yearly cycle of review based on the ISB HaSC review periods for existing standards, or for those areas requiring urgent resolution, on the basis of best endeavours resolution in a shorter timescale as required.

In terms of versioning, it is suggested that small incremental versions are released where a design fix has been provided due to a clinical incident. During each three yearly review, if there are significant changes to a standard based on evidence gathered proactively, then a more significant increment is probable. In all cases, changes to the standard will be reviewed by the ISB HaSC for approval prior to publishing.

Ownership

DHID will maintain ownership of the standards within the Technology Office, under the Data Standards and Products (DS&P) Directorate. For the duration of the CUI programme, the CAPS Project team within DS&P will provide the single point of contact in relation to any issues or amendments arising out of Live implementations in the NHS.

Once the CUI CAPS project is completed the Business As Usual (BAU) team within the Data Standards and Products Directorate will form the first point of contact for any questions and queries about CUI CAPS standards.

CUI CAPS Clinical Safety Management System

The following section outlines the *high level* CUI CAPS Clinical Safety Management System – which is predominantly based on the DHID Clinical Safety Management System and the emerging information standard, “Application of Clinical Risk Management to the Manufacture of Health Software.”

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Structure of the CUI CAPS Clinical Safety Team

The list below identifies the key people responsible for clinical safety within the CUI CAPS project. Although, all members of the CUI CAPS team are expected to integrate clinical safety into every aspect of development work. Appropriate awareness raising and education has been provided 'on site' by the current Clinical Safety Advisor. Relevant clinical team members of the CUI CAPS team (known as the Specific Audience) have attended the DHID training in human factors and risk assessment, in order to become "Accredited Clinicians" in Patient Safety Assessment (PSA) workshops and understand the required safety processes.

- Beverley Scott, Clinical Safety Advisor and Registered Mental Health Nurse (RMN) (Accredited Clinician) – February 2008 – September 2009.
- Lindsey Butler, Clinical Safety Advisor and Registered General Nurse (RGN) (Accredited Clinician) – from August 2009 onwards.
- Stephen Corbett, Head of User Interface Design, Technology Office.
- Tim Chearman, CUI CAPS Project Lead and User Experience Designer.
- Frank Cross, Consultant Surgeon and Senior Clinical Advisor (Accredited Clinician).
- Pete Johnson, Clinical Architect and General Practitioner (Accredited Clinician).

Due to the size of the CUI CAPS team, a internal clinical safety *committee* is not required. Instead the CAPS safety team as defined above meets on a regular basis with the design team to review the CUI CAPS clinical safety approach, individual Hazard Logs and clinical safety products in production and planned for future releases.

The CUI CAPS team utilise the safety assurance and sign off processes of the DHID Clinical Safety Group (CSG) and the Information Standards Board for Health and Social Care (ISB HaSC) for CUI design guidance.

Clinical Safety Products and Sign Off

The following clinical safety specific products will be produced for each CUI CAPS design guide:

- Hazard Log.
- Clinical Safety Case (including the Patient Safety Assessment).
- Safety Closure Report.

The DHID CSMS supports staged submission of the above documents to the Clinical Safety Group as a usual route to achieving Clinical Authority to Release (CATR). However, due to tight CUI CAPS project timescales, ie. typically 3 months per design

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guide, and limited Clinical Safety Advisor resource, the CUI CAPS Clinical Safety Cases and Safety Closure Reports will normally be compiled and submitted post design guide development.

Depending on the size and complexity of the design guide under consideration, project timescales and agreement from the CSG, the Clinical Safety Case and Safety Closure Report may be combined into one document and one submission to the CSG (as in this instance).

Whilst design guides may be published on the CUI website, www.cui.nhs.uk (registration required) before receiving CATR, both the safety and information standards status of each design guide is clearly identified on the website, and a caveat included in the document itself with respect to the patient safety process.

The latest safety information can be requested by suppliers by emailing the mailbox TechOffice.ClinicalSafety@nhs.net

Clinical Safety Risk Management File

All documents created as part of the CUI CAPS Clinical Safety Management System are maintained, version controlled and managed by the CUI CAPS Clinical Safety Advisor (CSA).

Documents are stored on the CUI CAPS Clinical Safety SharePoint site, which has access control measures in place. Members of the CSG or ISB HaSC can request access by contacting the CUI CAPS CSA.

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CUI CAPS Clinical Safety Approach

The CUI CAPS project has developed their clinical safety management approach based on the Clinical Safety Management System (CSMS) prescribed to system suppliers by the DHID and the emerging information standard, "Application of Clinical Risk Management to the Manufacture of Health Software."

Although outputs from the CUI CAPS project, namely text based user interface design guidance, are not systems per se, they will inform the user interface work carried out by clinical system suppliers which will in turn end up being part of a live deployed system. It is crucial, therefore, that the outputs from CUI CAPS are also clinically safe.

The DHID CSMS outlines four risk areas: End to end clinical process risk; Technical risk; Message risk; and Patient safety risk. As the CAPS project is focused on the user interface layer of an application, the scope of the CAPS safety process is limited to patient safety risks. The other risk types are classed out of CUI scope. However, where related risks are raised, CUI CAPS will record and transfer those risks to the appropriate parties.

As the CUI CAPS Project is responsible for developing user interface design guidelines which will be utilised within clinical applications - not the development of applications themselves – the CUI CAPS Clinical Safety Cases are developed as Pre-Operational. Taking a typical software development lifecycle (see the table below), the scope of CUI design guidance covers the stages of Define Requirements, Design and Develop. As such, the CUI CAPS Clinical Safety Cases will cover these stages also (although it important to note that the Patient Safety Assessment's themselves are very much focused upon context of use). It is the intention that the CUI CAPS safety documentation will be the starting point for NHS clinical application developers upon which to build their own Clinical Safety Case and, ultimately, pass on to end users within the live estate.

Manufacturers	Define Requirements Design Develop Build Test Release
End Users	Deploy Use Decommission

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In order to build the Clinical Safety Case, each CUI CAPS design guide will undergo an end to end Patient Safety Assessment (PSA). It is important to note that a Patient Safety Assessment (PSA) is not a distinct activity, scheduled into the development lifecycle as a milestone. A PSA is the term used to describe the end to end, continuous, iterative assessment to establish the safety integrity of the deliverable. It is integrated into the development lifecycle for each CAPS deliverable right from the outset.

A CUI CAPS PSA consists of two main 'processes': Hazard identification and risk assessment; and Hazard analysis and risk management. Various safety specific activities (eg. risk assessment workshop), methods (eg. Structured What If Technique, Hierarchical Task Analysis) and tools (eg. hazard checklists, cause and effect (Fishbone) diagrams) can be used during a PSA. However, all usual activities undertaken as part of normal research and development processes for CUI CAPS, ie. non safety specific, will pro-actively elicit patient safety issues and hazards on an ongoing basis. These are then fed into the master Hazard Log via the Clinical Safety Advisor for management.

As part of an end to end Patient Safety Assessment (PSA), the following **safety specific** activities should take place for each CUI CAPS deliverable (design guide):

- Each CAPS deliverable should have an assessment during the scoping phase to identify the minimum number and type of *safety specific* activities to take place and when.
- As a minimum, each CAPS deliverable should have:
 - A web based survey, including safety specific questions.
 - One to one interviews with representative end users, including safety specific questions.
 - Patient Safety Risk Assessment (PSA) workshops, eg.
 - 1 x Opening Risk Assessment workshop with representative end users (pre design: safety requirements and objectives and preliminary hazard identification and risk assessment).
 - 1 x Closing Risk Assessment workshop with representative end users (post design: risk identification, grading and management).
 - Interim Risk Assessments can take place as required, such as when a design guide spans multiple releases.
- The purpose of an Opening Risk Assessment is to inform the CUI CAPS requirements process (including safety requirements). Very often, the ORA can raise hazards and issues that are either out of scope or not patient safety specific – as it is conducted during the very early, conceptual stages of design.

Approved

- All hazards and issues, irrespective of where and when they are raised, go into the Hazard Log for central tracking by the CSA.
- The purpose of a Closing (or Interim) Risk Assessment is to formally risk assess final (or near final) designs, using a structured approach to identify hazards and grade and manage risks.
- Regular meetings take place with the CUI CAPS safety and design teams to review live hazards and manage to an acceptable level.
- **The number and type of activities, methods and tools varies across CUI CAPS deliverables. Ultimately, the overall aim of the PSA is to build a progressive Clinical Safety Case.**